Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2026

Course: B.Tech
Code: 105614

Subject: Web and Internet Technology

Time: 03 Hours
Full Marks: 70

Instructions: -

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) Consider the following statements about a Web 2.0 application:
(i) Users can generate and modify content
(ii) The application supports collaboration among users
(iii) Every interaction requires the server to send a completely new static HTML document
(iv) User participation is an important part of the application
Which statements are consistent with the Web 2.0 paradigm?
(i) 1 and 2 only
(ii) 1, 2 and 4 only
(iii) 2 and 3 only
(iv) 1, 2, 3 and 4

(b) Consider the following statements:
(i) HTML is primarily concerned with structuring/presenting web documents
(ii) XML allows user-defined tags
(iii) CSS is primarily used to define the presentation of HTML/XML content
(iv) XSL is associated with transforming/presenting XML data
Which combination is correct?
(i) 1 and 2 only
(ii) 1, 2 and 3 only
(iii) 1, 2, 3 and 4
(iv) 2 and 4 only

(c) A JavaScript program needs to:
(i) Open a new browser window
(ii) Determine the current URL
(iii) Navigate to another URL
Which browser-related object is most relevant?
(i) document only
(ii) XML Schema
(iii) DTD
(iv) BOM-related objects such as window and location

(d) Consider the following program:

```
<student roll="101">
  <name>Ravi</name>
</student>
```

Which statement is correct?
(i) roll is an attribute of the student element
(ii) roll is an element containing 101
(iii) 101 is an XML declaration
(iv) student is an attribute of roll

(e) An XML document contains thousands of elements. An application needs to identify a specific fragment within that document rather than merely identifying the complete document.Which technology is specifically intended for this purpose?
(i) XLink
(ii) DTD
(iii) XPointer
(iv) XSL

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(f) Which situation most strongly motivates using XML Schema rather than a basic DTD?

(i) The document requires only element names
(ii) The document requires richer data-type definitions
(iii) The document must contain HTML styling
(iv) The document must execute JavaScript

(g) Suppose a program needs to repeatedly navigate from a parent XML element to its child elements and modify different nodes in memory. Which model is more suitable?

(i) DOM
(ii) SAX
(iii) DTD
(iv) XPointer

(h) A web page retrieves updated XML data and modifies only a table on the page. The browser does not reload the entire HTML document. Which property of AJAX is being demonstrated?

(i) Asynchronous communication and partial page update
(ii) Static document rendering
(iii) XML validation
(iv) Client-side compilation

(i) Suppose an AJAX request has successfully received an XML response. Which sequence is most appropriate?

(i) Execute CSS → compile XML → send request
(ii) Update HTML → send response → access XML nodes
(iii) Modify DTD → reload browser → process response
(iv) Receive response → access XML nodes → process data → update HTML

(j) Consider an ASP application in which a server-side object has:
(i) Properties describing its state
(ii) Methods performing operations
Which statement is consistent with the ASP topics in the syllabus?

(i) ASP variables cannot participate in control structures
(ii) ASP objects can only represent CSS styles
(iii) ASP does not support database connectivity
(iv) Properties and methods provide ways to work with ASP objects

Q.2 (a) Explain N-tier software architecture with a suitable diagram and describe the role of each tier. [7]

(b) Describe the evolution of the World Wide Web from Web 1.0 to Web 2.0 with a suitable case study. Discuss the major characteristics, technologies and collaborative features of Web 2.0 applications. [7]

Q.3 (a) Design an HTML user registration form with an appropriate layout structure using the following form controls: [7]
(i) Name
(ii) Password and Confirm Password
(iii) Gender
(iv) Skills/Interests
(v) Country
(vi) Submit and Reset
Write the complete HTML code using the <form> tag and appropriate attributes such as action, method, name, value and type.

(b) Explain Cascading Style Sheets and discuss the Inline and External methods of integrating CSS with HTML. Illustrate each method with a suitable example. [7]

Q.4 (a) Describe the anatomy and basic structure of an XML document with a suitable example. [7]

(b) Explain XLink and XPointer in XML. Differentiate between simple and extended XLinks and state how XPointer is used to identify specific parts of an XML document. [7]

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Q.5 (a) Explain the importance of XML Schema and discuss its advantages over Document Type Definition. [7]

(b) Differentiate between Document Object Model and Simple API for XML parsers. Compare their memory usage, parsing mechanism (tree-based vs. event-driven), speed and suitable use cases. [7]

Q.6 (a) Write a JavaScript program to create and use a JavaScript object for storing student details such as name, roll number and course. Display the object properties dynamically on a web page. [7]

(b) Explain the XMLHttpRequest object and describe its request and response objects with a suitable example. [7]

Q.7 (a) Discuss the use of AJAX for creating dynamic HTML pages and highlight the role of CSS and XML in AJAX applications. [7]

(b) Explain the use of XML and the XMLHttpRequest object in implementing AJAX applications with a suitable example. [7]

Q.8 (a) Describe the basic structure of a PHP program and discuss the important features of PHP with a suitable example. [7]

(b) Explain database connectivity in PHP and describe how records can be retrieved from a MySQL database using a SELECT statement. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) URL
(b) JavaScript Events
(c) XML Validity
(d) AJAX Components

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